Project Summary

Target Model: TUSK E10T (Telescopic Pallet APR)			
Design Parameter	**Specification Value**		
Base Robot Model Reference	TUSK E10T (Telescopic extending forks & folding wheels)		
Target Pallet Compatibility	EUR-2 (1200x1000mm) & EUR-1 (1200x800mm) including double-sided pallets		
Maximum Payload Capacity	1000 kg (1.0 Ton)		
Robot Self-Weight	320 kg (including telescopic rails, carriage, and battery)		
Kinematic Layout	Differential Drive (2x Active mid-wheels, 4x Corner spring-suspended casters)		
Maximum Speed (Loaded / Unloaded)	1.5 m/s (Loaded) / 2.0 m/s (Unloaded)		
Lifting Height Range	75 mm (Lowered) to 175 mm (Fully Raised)		
Lifting Mechanism	Central vertical lifting carriage driven by a single motor and ball screw		
Fork Extension Mechanism	Synchronized telescopic guide rails driven by a single motor via a splined shaft		
Fork Wheel Mechanism	Passive retractable load rollers (rods pull rollers up when forks retract, and push them down during lift)		
Fork Width Adjustment Mechanism	Transverse Lead Screw for sliding forks on carriage (Adjustable span 550mm - 900mm)		
Power Source	48V 60Ah Lithium Iron Phosphate (LiFePO4) Battery with Smart BMS		
Navigation System	Hybrid SLAM (2D LiDAR + 3D Depth Camera) & DM QR-code navigation		
Control Hardware	ROS 2 Navigation PC (Intel Core i5) + STM32 Microcontroller (micro-ROS)		
Aisle Requirements	Minimum Travel Aisle: 1.4m | Minimum Pick Aisle: 2.0m		

Bill of Materials

BOM: TUSK E10T Telescopic Pallet-Handling AMR (1-Ton)									
Item ID	**Category**	**Component Name**	**Description**	**Specifications**	**Supplier / Manufacturer**	**Qty**	**Unit Cost ($)**	**Total Cost ($)**	**Engineering Notes**
AMR-BOM-001	Mechanical	Chassis Base Frame	Heavy-duty structural steel frame, laser-cut & welded plates	10mm A36 Steel Plates, Custom Weldment	Local CNC/Weld Shop	1	$650.00		Main frame holding battery, drive motors, and vertical guide rails
AMR-BOM-002	Mechanical	Vertical Lifting Carriage	Slide weldment plate that raises/lowers the entire fork assembly	8mm Steel Plates, Custom CNC Weldment	Local CNC/Weld Shop	1	$380.00		Rides vertically on chassis rails; holds the fork width and extension assemblies
AMR-BOM-003	Mechanical	Suspended Left & Right Forks	Low-profile sliding fork channels housing pull-rods	Forks thickness 65mm, High-Strength Steel Q345	Local CNC/Weld Shop	2	$450.00		Provides the support surface for lifting pallets
AMR-BOM-004	Mechanical	Fork Width Adjustment Slides	Linear rails & carriages to slide forks transversely on carriage	Hiwin HGR20 Rails (L=800mm) + HGW20CC Blocks	Hiwin / local distributor	2	$120.00		Allows adjusting fork span for EUR-1 and EUR-2 pallets
AMR-BOM-005	Mechanical	Fork Extension Telescopic Slides	Heavy-duty multi-stage telescopic linear rails for fork travel	Rollon DSS43-1170 Telescopic Rails, Over-extension	Rollon / local distributor	2	$380.00		Allows forks to extend 1400mm forward from the lift carriage
AMR-BOM-006	Mechanical	Fork Wheel Folding Linkages	Mechanical swingarms and long pull-rods running inside forks	Custom Q345 linkages + pinned joints	Local CNC/Weld Shop	2	$110.00		Passively folds load rollers up when forks retract, and down when lifting

AMR-BOM-007	Mechanical	Central Lifting Ball Screw	High-capacity vertical ball screw for carriage lift	SFU2505 Ball Screw (L=400mm), C7 Accuracy	TBI Motion / Misumi	1	$95.00		Single central screw to lift the entire carriage vertically
AMR-BOM-008	Mechanical	Vertical Guide Rails	Heavy-duty linear profile rails for carriage vertical guide	Hiwin HGR25 Rails (L=500mm) + HGW25CC Blocks	Hiwin / local distributor	2	$115.00		Mounted vertically on chassis front face for lift stability
AMR-BOM-009	Mechanical	Heavy-Duty Drive Wheels	Polyurethane tread wheels on cast iron core for drive traction	Dia 200mm, Width 75mm, Keyway hub	Blickle / local supplier	2	$75.00		Mid-drive wheels for differential steering layout
AMR-BOM-010	Mechanical	Heavy-Duty Caster Wheels	Swivel casters with polyurethane tires for chassis corners	Dia 100mm, load capacity 350kg each	Blickle / Misumi	4	$45.00		Four corner support wheels with spring suspension
AMR-BOM-011	Mechanical	Fork Tip Load Rollers	Low-profile tandem rollers inside fork tips, mounted on folding swingarms	Dia 60mm, polyurethane, needle bearings	Blickle / local supplier	4	$30.00		Sits under the front part of the forks, supports pallet weight
AMR-BOM-012	Mechanical	Bearing Block & Couplings	Flanged support bearings for ball screws and flexible jaw couplings	FK20/FF20 support blocks, D35 L50 Couplings	Misumi / local supplier	1	$70.00		Connects central lift motor to the vertical ball screw
AMR-BOM-013	Actuation	Differential Drive Motors	Brushless DC servo motors with planetary gearboxes & electromagnetic brakes	48V 750W, 3000RPM, 1:20 Planetary Gearbox, IP54	Leadshine / ZYT Motor	2	$380.00		Provides 1.5 m/s speed and high starting torque
AMR-BOM-014	Actuation	Central Lifting BLDC Motor	Brushless DC motor with planetary gearbox to drive vertical lift ball screw	48V 1000W, 3000RPM, 1:15 Planetary Gearbox, Braked	Leadshine / ZYT Motor	1	$420.00		Single motor to raise/lower the entire lifting carriage

AMR-BOM-015	Actuation	Fork Extension Geared Motor	Geared motor driving a transverse splined shaft for synchronized fork travel	24V 200W Geared DC Motor + Splined Drive Shaft	Leadshine / local supplier	1	$240.00		Single motor to extend/retract both forks simultaneously
AMR-BOM-016	Actuation	Fork Width Adjustment Motor	DC geared motor with integrated encoder for transverse lead screw drive	24V 60W, 100RPM, planetary gear	DFRobot / local supplier	1	$95.00		Adjusts the width slide of the forks dynamically
AMR-BOM-017	Control	Main Navigation Controller	Industrial PC running Linux Ubuntu and ROS 2 Navigation stack	Intel Core i5-1145G7, 16GB RAM, 256GB SSD, 9-36V DC	Neousys / Advantech	1	$750.00		Runs SLAM, Nav2, path planning, and fleet interface
AMR-BOM-018	Control	Low-Level MCU Board	Microcontroller board running micro-ROS for hardware interfaces	Teensy 4.1 or STM32F407 Board	STMicroelectronics / PJRC	1	$45.00		Handles encoder feedback, limit switches, and motor commands
AMR-BOM-019	Control	BLDC Motor Drivers	High-performance dual-channel CAN-bus brushless motor controllers	Roboteq SBL2360, 48V, Dual 60A per channel	Roboteq	2	$490.00		1 driver for 2x drive motors; 1 driver for 1x lift motor & 1x extension motor
AMR-BOM-020	Control	Smart Battery Pack	Lithium Iron Phosphate (LiFePO4) battery with RS485/CAN communication	48V 60Ah, 150A continuous discharge, built-in Smart BMS	Efest / RELiON / Custom	1	$1,200.00		Provides 8+ hours of operating runtime
AMR-BOM-021	Control	Power Distribution Board	Custom distribution board with circuit breakers, fuses, and relays	Custom PCB, heavy-copper traces, 100A main contactor	Local PCB shop	1	$150.00		Safely routes power to PC, drivers, and sensors

AMR-BOM-022	Control	Automatic Fast Charger	Industrial smart battery charger with docking pads for auto-charging	48V 15A output, contact pads for floor dock	Delta Q / local supplier	1	$350.00		Interfaces with floor docking plate for automated charging
AMR-BOM-023	Sensors	2D Navigation LiDARs	Safety laser scanners for SLAM and 360-degree obstacle detection	Sick TiM561 or Hokuyo UST-10LX, 10m range	Sick / Hokuyo	2	$980.00		Mounted diagonally (front-right, rear-left) for full coverage
AMR-BOM-024	Sensors	3D Obstacle Avoidance Cameras	Depth cameras for stereoscopic obstacle and height detection	Intel RealSense D435i or Orbbec Astra Pro	Intel / Orbbec	2	$220.00		Mounted front and rear for volumetric obstacle check
AMR-BOM-025	Sensors	Pallet Detection Sensors	Diffuse reflection photoelectric sensors in forks	Sick WL9 or Keyence PR-M, miniature photoeyes	Sick / Keyence	4	$85.00		Detects pallet presence and docking alignment in fork channels
AMR-BOM-026	Sensors	High-Precision IMU	Inertial Measurement Unit for SLAM odometry correction	9-axis IMU, Bosch BNO055 or Xsens MTi-3	Adafruit / Xsens	1	$75.00		Provides angular rate and linear acceleration data
AMR-BOM-027	Sensors	Inductive Limit Switches	Proximity sensors for lift height limits and fork width calibration	Omron E2E cylindrical proximity sensors	Omron	6	$25.00		End-stop detection for mechanical linkages
AMR-BOM-028	Sensors	Safety Bumpers & E-Stops	Pressure-sensitive contact edges and manual emergency stops	TapeSwitch bumper strip + 2x red mushroom E-stops	TapeSwitch / local supplier	1	$180.00		Physical safety overrides to stop motion immediately
AMR-BOM-029	Sensors	Warning Light Tower & Siren	Audible and visual status indicator beacon	RGB LED Tower, 85dB siren, 24V	Banner Engineering / local	1	$75.00		Signals robot state (idle, moving, fault, charging)
							GRAND TOTAL ESTIMATED HARDWARE COST		

Kinematics & Calculations

E10T Sizing & Engineering Calculations				
1. Central Vertical Lift Carriage Sizing				
Parameter / Step	**Symbol / Formula**	**Value**	**Unit**	**Explanation**
Total Load to Lift	*W = (Payload + Lift Carriage)*	**1080**	kg	1000kg max payload + 80kg carriage and forks weight
Total Lifting Weight Force	*F_g = W * 9.81*		N	Total gravitational force of load
Required Lifting Force	*F_lift = F_g*		N	Vertical force to lift the entire carriage (direct vertical lift)
Lifting Ball Screw Pitch	*p*	**5**	mm	Linear travel per rotation of the central vertical ball screw
Lifting Ball Screw Lead	*L = p / 1000*		m	Ball screw pitch in meters
Ball Screw Mechanical Efficiency	*eta_screw*	**0.90**	decimal	Frictional efficiency of rolling balls in screw threads
Screw Drive Torque Required	*T_screw = (F_lift * L) / (2 * pi * eta_screw)*		Nm	Torque required on the vertical ball screw shaft to lift load
Lifting Gearbox Reduction Ratio	*R_gearbox*	**15**	ratio	Gear ratio of planetary gearbox on lift motor
Gearbox Efficiency	*eta_gearbox*	**0.95**	decimal	Planetary gear stage efficiency
Peak Lift Motor Torque Required	*T_motor = T_screw / (R_gearbox * eta_gearbox)*		Nm	Required output torque from the single central BLDC motor before gearing
2. Differential Drive Traction Sizing				
Parameter / Step	**Symbol / Formula**	**Value**	**Unit**	**Explanation**
Total Gross Mass (AMR + Payload)	*M = M_amr + M_payload*	**1320**	kg	Robot mass 320kg (E10T weight) + 1000kg payload

Maximum Travel Speed	*v_max*	**1.50**	m/s	Design speed of AMR at full capacity
Acceleration Time to Max Speed	*t_acc*	**2**	s	Time limit to reach full speed
Required Linear Acceleration	*a = v_max / t_acc*		m/s^2	Target acceleration rate
Acceleration Force Required	*F_acc = M * a*		N	Force needed to accelerate mass
Rolling Resistance Coefficient	*f_r*	**0.02**	decimal	Polyurethane wheels on smooth, hard industrial concrete floor
Rolling Resistance Force	*F_roll = M * g * f_r*		N	Force to overcome rolling friction
Total Tractive Force	*F_total = F_acc + F_roll*		N	Sum of acceleration and rolling friction forces
Number of Driving Motors/Wheels	*N_drive*	**2**	qty	Differential drive layout has two independent driven wheels
Tractive Force per Drive Wheel	*F_wheel = F_total / N_drive*		N	Tractive force share per driving wheel
Drive Wheel Radius	*R_wheel*	**0.10**	m	100mm wheel radius (200mm diameter drive wheel)
Peak Wheel Torque Required	*T_wheel = F_wheel * R_wheel*		Nm	Required torque at the wheel axle
Drive Gearbox Reduction Ratio	*R_drive_gearbox*	**20**	ratio	Planetary gearbox reduction ratio
Drive Gearbox Mechanical Efficiency	*eta_drive_box*	**0.95**	decimal	Efficiency of planetary stages
Peak Drive Motor Torque Required	*T_drive_motor = T_wheel / (R_drive_gearbox * eta_drive_box)*		Nm	Required motor electromagnetic output torque
Rated Motor Rotation Speed	*omega_motor = (v_max / R_wheel) * R_drive_gearbox * 60 / (2*pi)*		RPM	Rotational speed of motor shaft at max linear velocity
Mechanical Power Output per Motor	*P_motor = T_drive_motor * (omega_motor * 2*pi / 60)*		W	Continuous mechanical wattage required per motor

Wiring Architecture

E10T Electrical Connections & Signal Routing					
Connection ID	**Source Device**	**Destination Device**	**Cable Type**	**Interface Protocol**	**Function / Description**
PWR-01	Smart Battery Pack (48V)	Main Circuit Breaker / Switch	AWG 4 Silicon Wire	DC Power	Main power input connection
PWR-02	Circuit Breaker	Power Distribution Board	AWG 6 Silicon Wire	DC Power	Fused main DC distribution bus
PWR-03	Power Distribution Board	Drive Motor Drivers (Roboteq)	AWG 10 Silicon Wire	DC Power	High current feed for drive motors (fused)
PWR-04	Power Distribution Board	Lifting Motor Drivers (Roboteq)	AWG 10 Silicon Wire	DC Power	High current feed for lifting motors (fused)
PWR-05	Power Distribution Board	Voltage Regulator (48V to 12V/24V)	AWG 16 wire	DC Power	Power feed to step-down converters
PWR-06	Voltage Regulator (24V)	Industrial PC (Nav Controller)	AWG 18 wire	24V DC Power	Regulated logic power for IPC
PWR-07	Voltage Regulator (24V)	Sensors & Low-level Controllers	AWG 20 wire	24V/12V Power	Power for LiDARs, cameras, and MCU
SIG-01	2D Navigation LiDARs (2x)	Industrial PC	Cat6 Shielded RJ45	Ethernet (TCP/IP)	Point cloud data transmission for SLAM
SIG-02	3D Obstacle Cameras (2x)	Industrial PC	USB 3.0 Type-C Cable	USB 3.0	RGB-D frames for stereoscopic obstacle checks
SIG-03	IMU Sensor	Low-Level MCU Board	4-pin Ribbon Cable	I2C / UART	Raw accelerometer/gyroscope measurements
SIG-04	Wheel Encoders (2x)	Low-Level MCU Board	Shielded 6-wire Cable	Quadrature (A/B)	Incremental drive wheel feedback for odometry

SIG-05	Lift Linear Encoder	Low-Level MCU Board	Shielded 6-wire Cable	SSI / SSI-SPI	Absolute vertical position of the lifting carriage
SIG-06	Limit & Proximity Switches (6x)	Low-Level MCU Board	3-wire Sensor Cable	Digital I/O (24V)	End-of-travel safety flags for lift and slide
SIG-07	Pallet Photoelectric Sensors (4x)	Low-Level MCU Board	3-wire Sensor Cable	Digital I/O (24V)	Detects pallet legs and correct insertion
COM-01	Industrial PC	Low-Level MCU Board	Shielded USB Cable	micro-ROS Serial (USB)	Command velocities (cmd_vel) and odometry feedback
COM-02	Low-Level MCU Board	Drive Motor Drivers	Shielded twisted pair	CAN-bus	Position/velocity control frames for wheels
COM-03	Low-Level MCU Board	Lifting Motor Driver	Shielded twisted pair	CAN-bus	Vertical lift motion control
COM-04	Smart Battery BMS	Industrial PC	RS485 to USB converter	Modbus RTU / CAN	State of Charge (SoC), voltage, and diagnostic logs